Appendix

Tingchang Wang, Zhigang Li

1 Form of the estimating equations

As proposed in (Preisser et al. (2002)), the weighted generalized estimating equation takes the form:

$$\sum_{i=1}^{N} D(Z_i, \alpha)^T V_i^{-1} W_i (U_i - \mu(Z_i, \alpha)) = 0$$

Since the estimators' consistency and other asymptotic properties doesn't depend on the specification correlation, we use independent working correlation strcture to develop the equation. The components in equation above can be further specified as:

$$\begin{aligned}
U_i &= \left(log(Y_i^1 / Y_i^{K+1}), \cdots, log(Y_i^K / Y_i^{K+1}) \right)^T \\
\mu(Z_i, \alpha) &= \left(E[log(Y_i^1 / Y_i^{K+1})], \cdots, E[log(Y_i^K / Y_i^{K+1})] \right)^T \\
&= (Z_i^T \alpha^{1,K+1}, \cdots, Z_i^T \alpha^{K,K+1})^T \\
D(Z_i, \alpha) &= \frac{\partial \mu(Z_i, \alpha)}{\partial \alpha} \\
&= (Z_i^T, \cdots, Z_i^T) \\
V_i &= var(U_i) \\
&= diag(\phi_{1,K+1}, \cdots, \phi_{K,K+1}) \\
W_i &= diag(R_i^1 R_i^{K+1} w_{i1}, \cdots, R_i^K R_i^{K+1} w_{iK})
\end{aligned}$$

where $w_{ik} = 1/P(R_i^k = 1, R_i^{K+1} = 1 | Y_i^k, Y_i^{K+1})$ is the inverse probability of Y_i^k and Y_i^{K+1} being observed simultaneously for each $k = 1, \cdots, K$.

By evaluating the matrix operations, we obtain the form of equation 4) in section (4.1):

$$\sum_{i=1}^{N}\sum_{k=1}^{K}\frac{R_i^k R_i^{K+1}\phi^{-1}_{k,K+1}}{P(R_i^k=1,R_i^{K+1}=1|Y_i^k,Y_i^{K+1})}Z_i\big(\log(Y_i^k/Y_i^{K+1})-Z_i^T\alpha^{k,K+1}\big)=0, \tag{1}$$

Next, we show that equation (5) agrees with the general form of the estimating equation if all zeros are mechanical zeros.

Proof. For each sample i and each set of $\{k_1,k_2\}$, we have

$$\begin{aligned}
&E\left(\frac{R_i^{k_1}R_i^{k_2}\big(A^{k_1k_2}\mathcal{Z}_i\big)^T\phi^{-1}_{k_1k_2}\Big(U_i^{k_1k_2}-A^{k_1k_2}\mathcal{Z}_i\alpha^{K+1}\Big)}{P(R_i^{k_1}=1,R_i^{k_2}=1|Y_i^{k_1},Y_i^{k_2})}\right)\\
&=E\left(E\left(\frac{R_i^{k_1}R_i^{k_2}\big(A^{k_1k_2}\mathcal{Z}_i\big)^T\phi^{-1}_{k_1k_2}\Big(U_i^{k_1k_2}-A^{k_1k_2}\mathcal{Z}_i\alpha^{K+1}\Big)}{P(R_i^{k_1}=1,R_i^{k_2}=1|Y_i^{k_1},Y_i^{k_2})}\;\Big|\;Y_i^{k_1},Y_i^{k_2}\right)\right)\\
&=E\left(\frac{E\Big(R_i^{k_1}R_i^{k_2}\mid Y_i^{k_1},Y_i^{k_2}\Big)\big(A^{k_1k_2}\mathcal{Z}_i\big)^T\phi^{-1}_{k_1k_2}\Big(U_i^{k_1k_2}-A^{k_1k_2}\mathcal{Z}_i\alpha^{K+1}\Big)}{P(R_i^{k_1}=1,R_i^{k_2}=1|Y_i^{k_1},Y_i^{k_2})}\right)\\
&=E\left(\frac{P(R_i^{k_1}=1,R_i^{k_2}=1|Y_i^{k_1},Y_i^{k_2})\big(A^{k_1k_2}\mathcal{Z}_i\big)^T\phi^{-1}_{k_1k_2}\Big(U_i^{k_1k_2}-A^{k_1k_2}\mathcal{Z}_i\alpha^{K+1}\Big)}{P(R_i^{k_1}=1,R_i^{k_2}=1|Y_i^{k_1},Y_i^{k_2})}\right)\\
&=E\left(\big(A^{k_1k_2}\mathcal{Z}_i\big)^T\phi^{-1}_{k_1k_2}\Big(U_i^{k_1k_2}-A^{k_1k_2}\mathcal{Z}_i\alpha^{K+1}\Big)\right)\\
&=\big(A^{k_1k_2}\mathcal{Z}_i\big)^T\phi^{-1}_{k_1k_2}\Big(E\big(U_i^{k_1k_2}\big)-A^{k_1k_2}\mathcal{Z}_i\alpha^{K+1}\Big).
\end{aligned}$$

Because $U_i^{k_1k_2}=A^{k_1k_2}\Big(\log\big(Y_i^1/Y_i^{K+1}\big),\ldots,\log\big(Y_i^K/Y_i^{K+1}\big)\Big)^T$ and $E\big(\log(Y_i^k/Y_i^{K+1})\big)=$

$Z_i^T \alpha^{k,K+1}$ for all $k \in \{1, \ldots, K\}$ as shown in equation (3), we have

$$
\begin{aligned}
& E\big(U_i^{k_1 k_2}\big) - A^{k_1 k_2} \mathcal{Z}_i \alpha^{K+1} \\
&= A^{k_1 k_2} E\Big(\log\big(Y_i^1 / Y_i^{K+1}\big), \ldots, \log\big(Y_i^K / Y_i^{K+1}\big)\Big)^T - A^{k_1 k_2} \mathcal{Z}_i \alpha^{K+1} \\
&= A^{k_1 k_2} \left(E\Big(\log\big(Y_i^1 / Y_i^{K+1}\big)\Big), \ldots, E\Big(\log\big(Y_i^K / Y_i^{K+1}\big)\Big)\right)^T - A^{k_1 k_2} \mathcal{Z}_i \alpha^{K+1} \\
&= A^{k_1 k_2} \big(Z_i^T \alpha^{1,K+1}, \ldots, Z_i^T \alpha^{K,K+1}\big)^T - A^{k_1 k_2} \mathcal{Z}_i \alpha^{K+1} \\
&= A^{k_1 k_2} \mathcal{Z}_i \alpha^{K+1} - A^{k_1 k_2} \mathcal{Z}_i \alpha^{K+1} \\
&= \mathbf{0}.
\end{aligned}
$$

Thus equation (5) is a weighted estimating equation. □

References

Preisser, J. S., Lohman, K. K., and Rathouz, P. J. (2002). Performance of weighted estimating equations for longitudinal binary data with drop-outs missing at random. *Statistics in Medicine*, 21:3035–3054.